HR ANALYTICS -PREDICTING EMPLOYEE ATTRITION

INTRODUCTION

Employee attrition is one of the major challenges faced by organizations, as it impacts productivity, increases hiring costs, and affects overall workforce stability. By analyzing employee-related factors, organizations can identify the reasons behind attrition and take proactive measures to improve employee retention. This project focuses on analyzing HR data and predicting employee attrition using data analytics and machine learning techniques.

ABSTRACT

This project aims to analyze employee attrition patterns and identify the factors that contribute to employee turnover. Data preprocessing was performed using Python, and Exploratory Data Analysis (EDA) was conducted to uncover trends related to employee demographics, job roles, income levels, and work conditions. The processed dataset was then used to build a predictive model for attrition analysis. An interactive Power BI dashboard was developed to visualize key insights and support HR decision-making.

TOOLS USED

- Python (Pandas, NumPy, Seaborn, Matplotlib)
- Scikit-learn
- Power BI
- Jupiter Notebook
- Processed HR Attrition Dataset (CSV)

STEPS INVOLVED IN BUILDING THE PROJECT

1. Imported the HR Employee Attrition dataset into Python.

2. Cleaned and preprocessed the data by handling categorical variables and preparing the dataset for analysis.

3. Performed Exploratory Data Analysis (EDA) to understand employee attrition trends.

4. Analyzed important factors such as Age, Department, Job Role, Monthly Income, Overtime, and Years at Company.

5. Generated a processed CSV file for further analysis and visualization.

6. Built a machine learning classification model to predict employee attrition.

7. Evaluated model performance using appropriate metrics.

8. Imported the processed dataset into Power BI.

9. Created interactive dashboards and visualizations to identify attrition patterns and business insights.

10. Derived recommendations to help organizations reduce employee turnover.

Key Insights

1. Employees working overtime show a higher attrition rate compared to employees who do not work overtime.

2. Attrition is more common among employees with lower monthly income, indicating that compensation plays an important role in employee retention.

3. Employees with fewer years at the company have a higher tendency to leave the organization

4. Certain departments and job roles experience higher attrition levels, suggesting the need for department-specific retention strategies.

5. Younger employees show relatively higher attrition compared to senior employees.

6. Employee satisfaction, work-life balance, and career growth opportunities significantly influence retention.

7. The Power BI dashboard enables HR teams to monitor attrition trends and identify employee groups with higher risk of leaving the organization.

CONCLUSION

The project successfully analyzed employee attrition and identified key factors influencing employee turnover. Data preprocessing and predictive modeling were carried out using Python, while Power BI was used to create an interactive dashboard for effective visualization. The findings can help HR teams understand attrition trends, improve employee satisfaction, and implement strategies to increase employee retention.